

FREE ELECTRON THEORY OF METALS

Introduction:

The study of behavior of free electrons and various properties such as electrical, thermal and magnetic properties etc., in a solid is known as free electron theory of solids.

The electron theory of solids has been developed in three main stages.

(i). The classical free electron theory:

- This theory was developed by Drude and Lorentz in 1900.
- According to this theory, metals consist of positive ion cores and free electrons.
- The free electrons bound to move within the metal in a completely uniform potential field among these ion cores and obey the laws of classical mechanics.

(ii). The Quantum free electron theory:

- This theory was developed by Sommerfeld in 1928.
- The free electrons bound to move within the metal in a completely uniform potential field among the ion cores and obey quantum laws.

(iii). The Zone theory:

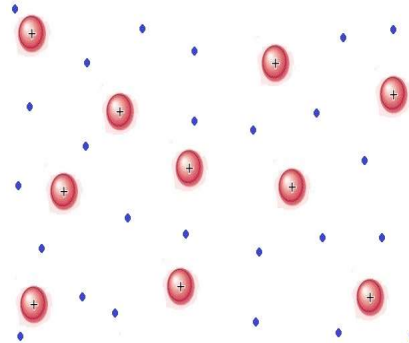
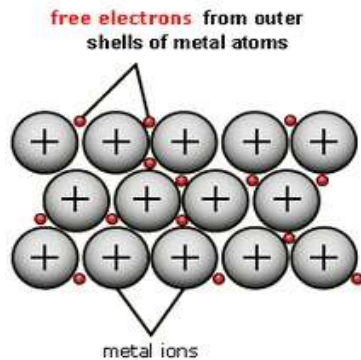
- Bloch stated this theory in 1928.
- According to this theory, the free electrons move in a periodic field provided by the lattice and obey quantum laws. This theory is also called “Band theory of solids”.

Classical free electron theory (Free electron gas model):

After discovery of electron by J.J.Thomson in 1897, Paul Drude developed the free electron theory of metals in 1900 and later it was extended by Hendrik Antoon Lorentz.

- According to this theory, metals consist of positive ion cores and free electrons.
- The free electrons bound to move within the metal in a completely uniform potential field among these ion cores.
- The behavior of free electrons moving within the metal is considered to be similar to that of atoms or molecules in perfect gas. These electrons are, therefore, also referred to as free electron gas and the theory is accordingly named as free electron gas model.
- The movement of free electrons obeys the laws of classical kinetic theory of gases.
- These free electrons move in random directions and collide with either positive ions fixed to the lattice or free electrons. All collisions are elastic, i.e., there is no loss of energy.
- The electrostatic force of attraction between the free electrons and the metallic ions are neglected.

- The mutual repulsion (electrostatic force of repulsion) among the free electrons is ignored, so that they move in all the directions with all possible velocities and obey the Maxwell-Boltzmann distribution of velocities.



Electrical conduction:

Consider a metallic conductor of length l and area of cross-section A .

Let n be the number of free electrons per unit volume and e be the charge of an electron.

Case 1:

When there is no electric field, all free electrons move in randomly in all directions. So the net current flow is zero.

Case2:

When an electric field is applied to the metallic conductor, it exerts a force on the free electrons (Lorentz force) which start accelerating towards the positive terminal.

As the free electrons move, they collide again and again with positive ions of the metal. Each collision destroys the extra velocity gained by the free electrons.

The average distance travelled by a free electron between two successive collisions is called mean free path (λ).

The average time interval between two successive collisions is called the relaxation time (τ). Its value is of the order of 10^{-14} second.

Although the free electrons are continuously accelerated by the electric field, collisions prevent their velocity becoming large. Finally free electrons acquire a constant average velocity is called drift velocity (V_d).

The constant average velocity acquired by the free electrons in a conductor subjected to an electric field is called drift velocity (V_d).

The drift velocity of free electron is of the order of 10^{-5} m/s.

Under an electric field:

The electron gains acceleration $a = \frac{\text{Drift velocity } (V_d)}{\text{Relaxation time } (\tau)}$

$$\therefore \text{The drift velocity } V_d = a\tau \longrightarrow (1)$$

$$\text{Each electron experiences a Lorentz force } F_L = eE \longrightarrow (2)$$

From Newton's Second law of motion,

$$\text{The force on the electron } F = ma \longrightarrow (3)$$

Equating equation (2) and equation (3), we have

$$ma = e E$$

$$a = \frac{e E}{m} \longrightarrow (4)$$

Substituting equation (4) in (1), we get

$$V_d = \left(\frac{e E}{m}\right) \tau \longrightarrow (5)$$

The current flowing through a conductor $I = neAV_d \longrightarrow (6)$

Substituting equation (5) in (6), we get

$$\begin{aligned} &= neA\left(\frac{e E}{m}\right) \tau \\ &= nA\left(\frac{e^2 E}{m}\right) \tau \longrightarrow (7) \end{aligned}$$

The current density $J = \frac{I}{A} \longrightarrow (8)$

Substituting equation (7) in (8), we get

$$\begin{aligned} &= \frac{nA\left(\frac{e^2 E}{m}\right) \tau}{A} \\ &= n\left(\frac{e^2 E}{m}\right) \tau \longrightarrow (9) \end{aligned}$$

The electrical conductivity $\sigma = \frac{J}{E} \longrightarrow (10)$

Substituting equation (9) in (10), we get

$$= \frac{n\left(\frac{e^2 E}{m}\right) \tau}{E}$$

According to kinetic theory of gases,

Relaxation time $\tau = \frac{\lambda}{\bar{c}}$ and

Root mean square velocity $\bar{c} = \sqrt{\frac{3K_B T}{m}}$

$$\therefore \text{The electrical conductivity } \sigma = n\left(\frac{e^2}{m}\right) \tau = n\left(\frac{e^2}{m}\right) \frac{\lambda}{\bar{c}} = \frac{ne^2 \lambda}{\sqrt{3mK_B T}}$$

(OR)

$$\text{The electrical resistivity } \rho = \frac{1}{\sigma} = \frac{m}{ne^2 \tau} = \frac{m \bar{c}}{ne^2 \lambda} = \frac{\sqrt{3mK_B T}}{ne^2 \lambda}$$

The above equations represent the electric conductivity and resistivity of metals.

Merits of classical free electron theory:

- It verifies Ohm's law.
- It explains the electrical conductivity of metals.
- It explains the electrical resistivity of metals.
- It explains the thermal conductivities of metals.
- It derives Wiedemanns-Franz law. (i.e., the relation between electrical conductivity and thermal conductivity)
- It explains optical properties of metals.

Demerits of classical free electron theory:

- This theory fails to explain the photo electric effect, Compton Effect and the black body radiation etc.
- This theory fails to explain specific heat of metals.
- According to this theory, the electrical conductivity (σ) of a metal is inversely proportional to its square root of its absolute temperature (T). i.e., $\sigma \propto \frac{1}{\sqrt{T}}$.
But practically, it has been observed that the electrical conductivity is inversely proportional its absolute temperature. i.e., $\sigma \propto \frac{1}{T}$.
Thus, this theory failed to explain electrical conductivity of metals.
- This theory fails to explain mean free path of electrons. Because theoretical and experimental values of mean free path (λ) does not coincide.

Quantum free electron theory:

- This theory was developed by Sommerfeld in 1928 based on the Pauli Exclusion Principle and Fermi- Dirac Statistics.
- According to this theory, metals consist of positive ion cores and free electrons.
- The free electrons bound to move within the metal in a completely uniform potential field among these ion cores and obey quantum laws.
- Mutual attraction between electrons and positive ions and the repulsion between individual electrons may be ignored.
- The Free electrons will have discrete energy states (or) levels in a metal.
- The occupation (or) distribution of electrons among these energy levels will be governed by Pauli's exclusion principle.
- According to Pauli's exclusion principles, no two electrons can have same set of quantum numbers in a given energy state. i.e., no two electrons can occupy the same energy level and each energy can accept only two electrons one with spin up ($\uparrow$) and other with spin down ($\downarrow$).
- At absolute zero temperature (0K), the first electron can occupy the lowest available energy state (E_0) with spin up ($\uparrow$) and the next electron can also occupy the same energy level with opposite spin down ($\downarrow$) of first electron. The third electron cannot occupy in this energy level, because of Pauli's exclusion principle and it will go to the

next energy (E_1). In this way, all free electrons can be occupied in the different energy states in the one dimensional potential box as shown in figure.

- The highest energy level occupied by electrons at absolute zero temperature is known as “Fermi energy level” (E_F). Above Fermi energy level all energy levels are empty. i.e., at 0K all the energy levels below Fermi energy level are filled by free electrons. So they possess different energies and velocities or momentums.
- The different velocities of these free electrons of a metal can be represented in velocity space as shown fig.
- At 0K, free electrons present in Fermi energy level possess maximum velocity represented by V_F .
- We assume a sphere of radius V_F , at the origin of velocity space as shown in fig.
- At 0K, all points inside the Fermi sphere are occupied.
- Each point inside the sphere from the origin represents the velocity of a free electron.
- The vectors joining different points inside the sphere from the origin represent velocity vectors.
- In the absence of electric field, the velocity vectors in the Fermi sphere present from origin will be cancelled with each other and the net velocity of electrons in all directions is zero.
- If the electric field E is applied on the free electrons along positive X-direction, then electrons experience a Lorentz force F_L along negative X-direction.
- The electrons which are present near the Fermi surface take electrical energy and occupy higher vacant energy levels as a result the sphere slightly displaces as shown fig. The remaining electrons unable to occupy higher vacant energy levels because Lorentz force is less on these electrons.

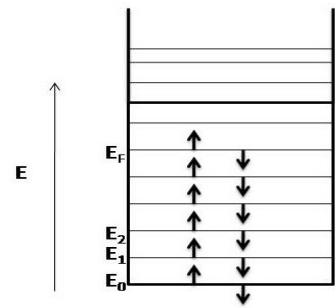


Fig: Distribution of free electrons in various energy levels at 0K.

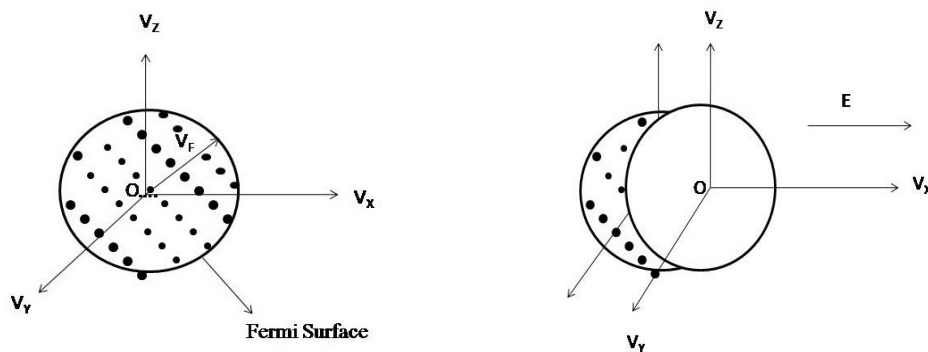


Fig: Displacement of Fermi sphere under applied electric field.

- Under an electric field:

Each electron experiences a Lorentz force $F_L = e E \longrightarrow (1)$

From Newton's Second law of motion,

The force on the electron $F = ma \longrightarrow (2)$

Equating equation (2) and equation (3), we have

$$ma = e E \longrightarrow (3)$$

The relation between momentum P and the wave or propagation vector K is given by

$$P = \hbar k \longrightarrow (4)$$

$$\text{But } P = mv \longrightarrow (5)$$

Equating equation (4) and equation (5), we have

$$mv = \hbar k$$

$$v = \frac{\hbar k}{m} \longrightarrow (6)$$

Differentiating equation (6) with respect to time gives acceleration,

$$a = \frac{dv}{dt} = \frac{\hbar}{m} \frac{dk}{dt} \longrightarrow (7)$$

Substituting equation (7) in (3), we get

$$m \frac{\hbar}{m} \frac{dk}{dt} = e E \longrightarrow (8)$$

$$dk = \frac{eE}{\hbar} dt \longrightarrow (9)$$

Integrating the equation (9), taking limits from 0 to t ;

$$\int_0^t dk = \int_0^t \frac{eE}{\hbar} dt$$

$$K(t) - K(0) = \frac{eEt}{\hbar} \longrightarrow (10)$$

Let the mean collision time and mean free path of electron present at Fermi surface is represented as τ_F and λ_F respectively then, we have:

$$\tau_F = \frac{\lambda_F}{v_F} \longrightarrow (11)$$

For an electron at Fermi surface, $t = \tau_F$ and $K(t) - K(0) = \Delta k$.

$$\text{Then } \Delta k = \frac{eE\tau_F}{\hbar} \longrightarrow (12)$$

Substituting equation (11) in (12), we get

$$\Delta k = \frac{eE}{\hbar} \left[\frac{\lambda_F}{v_F} \right] \longrightarrow (13)$$

If the increased velocity is Δv due to application of electric field then the current density

$$J = ne\Delta v \longrightarrow (14)$$

Where n is the number of free electrons per unit volume and e is the charge of the electron. From equations (6) and (13), analytically Δv can be expressed as

$$\Delta v = \frac{\hbar \Delta k}{m^*} = \frac{\hbar}{m^*} \frac{eE}{\hbar} \left[\frac{\lambda_F}{V_F} \right] = \frac{eE}{m^*} \left[\frac{\lambda_F}{V_F} \right] \longrightarrow (15)$$

Where m^* is the effective mass of free electron.

Substituting equation (15) in (14), we get

$$\begin{aligned} J &= ne \times \frac{eE}{m^*} \left[\frac{\lambda_F}{V_F} \right] \\ &= \frac{ne^2 E}{m^*} \left[\frac{\lambda_F}{V_F} \right] \longrightarrow (16) \end{aligned}$$

The electrical conductivity $\sigma = \frac{J}{E} \longrightarrow (17)$

Substituting equation (16) in (17), we get

$$\begin{aligned} \sigma &= \frac{\frac{ne^2 E}{m^*} \left[\frac{\lambda_F}{V_F} \right]}{E} \\ &= \frac{ne^2}{m^*} \left[\frac{\lambda_F}{V_F} \right] \quad \text{[From eq(11)]} \end{aligned}$$

$$\sigma = \frac{ne^2 \tau_F}{m^*}$$

This is the expression for electrical conductivity which is found to be in good agreement with experimental values.

Thus we can say that, according to this theory, all free electrons are not responsible for electrical conduction, only those free electrons which are very near to Fermi surface.

Merits of Quantum free electron theory:

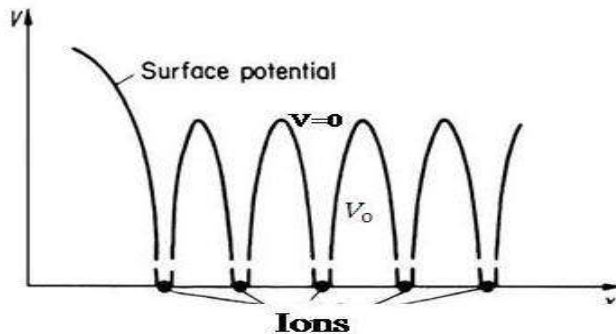
- It explains the electrical conductivity of metals.
- It explains the electrical resistivity of metals.
- It explains the thermal conductivities of metals.
- It derives Wiedemanns-Franz law. (i.e., the relation between electrical conductivity and thermal conductivity)
- It explains optical properties of metals.
- It successfully explains the photo electric effect, Compton Effect and the black body radiation etc.
- It explains the specific heat of metals.

Demerits of Quantum free electron theory:

- This theory fails to explain the differences between insulators, conductors and semiconductors.
- Positive hall coefficient of metals and
- Lower conductivities of divalent metals than monovalent metals.

Band theory or zone theory of solids:

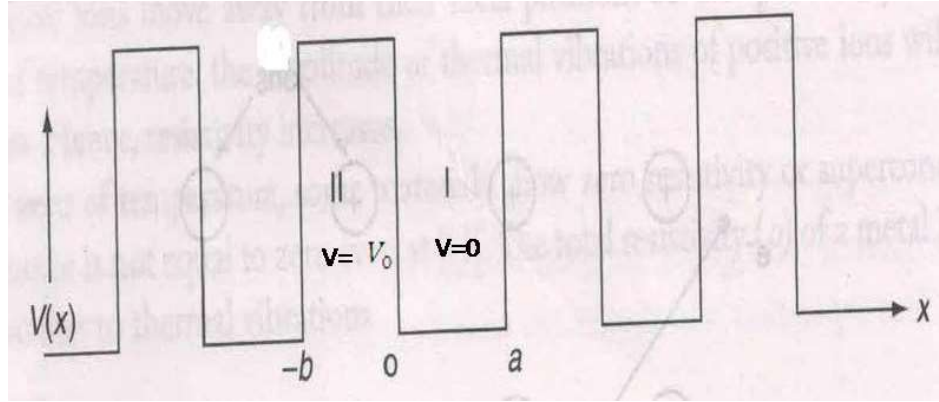
- This theory was introduced by Bloch in 1928.
- According to this theory, the free electrons move within a metal with periodic potential field provided by the positive ion cores and obey quantum laws.
- The potential of the electron varies periodically with periodicity of positive ion cores.
- The potential energy of energy is zero at the positive ion sites and maximum in between two ions as shown in fig.



- The periodic potential of electron is given by $V(x) = V(x+a)$.
- The Schrödinger wave equation for periodic potential is
- $$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - V(x)) \psi = 0 \longrightarrow (1)$$
- The solution of the above equation is
 - $\psi = \exp^{\pm ikx} u_k(x) \longrightarrow (2)$
 - Where $u_k(x)$ represents the periodic function and is given by $u_k(x) = u_k(x+a)$.
- But, it is very difficult to solve Schrödinger equation with sinusoidal potentials.
- So, Kronig and Penny modified the above curves as rectangular wells and barriers as shown in fig. This model is known as Kronig–Penny model.

Kronig–Penny model:

- According to this theory, the free electrons move within a metal with periodic potential field provided by the positive ion cores and obey quantum laws.
- The potential of the electron varies periodically with periodicity of positive ion cores.
- Kronig and Penny modified the periodic potential curves as rectangular wells and barriers as shown in fig.
- The potential energy of energy is zero at the positive ion sites and maximum in between two ions.



- From fig , we consider two regions : (i) Region I (Well)
(ii) Region II (Barrier)

Region I (Well):

In this region, the potential of electron is $V(x) = 0$ when $0 < x < a$.

The time independent Schrödinger wave equation for the region I (Well) is

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2mE}{h^2} \psi = 0 \longrightarrow (1)$$

$$\text{(or)} \frac{d^2\psi}{dx^2} + \alpha^2 \psi = 0 \longrightarrow (2)$$

$$\text{Where } \alpha^2 = \frac{8\pi^2mE}{h^2} \longrightarrow (3)$$

Region II (Barrier):

In this region, the potential of electron is $V(x) = V_0$ when $-b < x < a$.

The time independent Schrödinger wave equation for the region II (Barrier) is

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - V_0) \psi = 0 \longrightarrow (4)$$

$$\text{(or)} \frac{d^2\psi}{dx^2} + \beta^2 \psi = 0 \longrightarrow (5)$$

$$\text{Where } \beta^2 = \frac{8\pi^2mE}{h^2} \longrightarrow (6)$$

The solution for the equations (2) and (5) can be written in the Bloch's form as

$$\psi = \exp^{\pm ikx} u_k(x) \longrightarrow (7)$$

Where $u_k(x)$ represents the periodic function and is given by $u_k(x) = u_k(x+a)$ and $k = \frac{2\pi}{\lambda}$ is called propagation vector or wave vector.

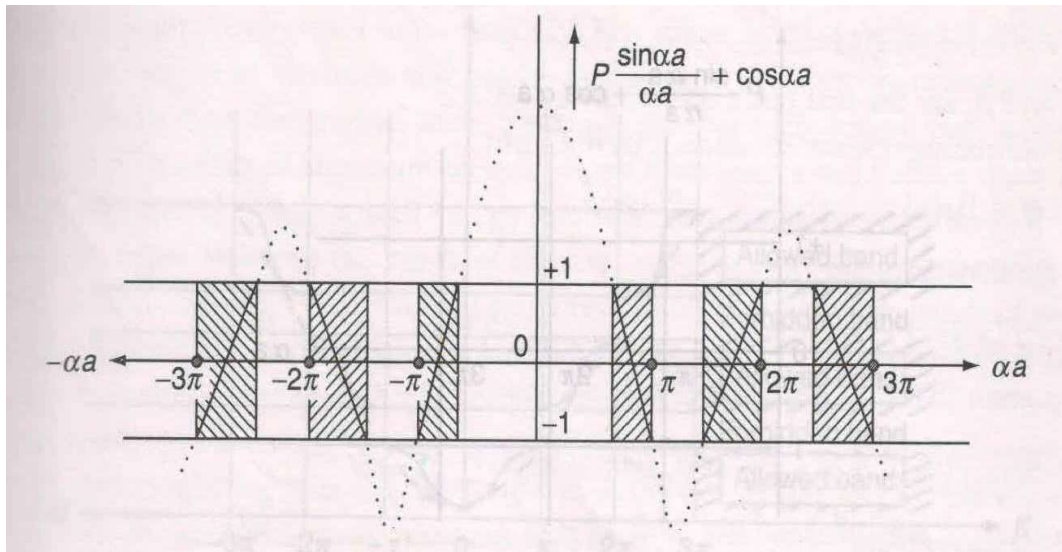
Differentiating equation (7) and substituting it in equations (2) and (5) and then further solving applying boundary conditions, finally we get

$$\left(\frac{mV_0ab}{h^2}\right) \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka \longrightarrow (8)$$

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka \longrightarrow (9)$$

Where $P = \frac{mV_0 ab}{\hbar^2}$ is known as scattering power of potential barrier and is measure of potential barrier strength with which the electrons are attracted by the positive ions.

In equation (9), there are only two variables α and k .



From fig (2) the following conclusions can be drawn:

- The energy spectrum of the electron consists of a number of allowed and forbidden energy bands.
- The width of the allowed energy band increases with increase of energy values i.e., increasing the values of αa .
- The width of the allowed energy band decreases with increase in the value of 'P' (potential barrier strength) i.e., with increasing in the binding energy of the electrons. As $P \rightarrow \infty$, the allowed bands are compressed into narrow lines i.e., energy levels and the energy spectrum becomes a line spectrum as shown in fig. If $P \rightarrow \infty$, then equation (9) has the only solution,

$$\text{i.e., } \sin \alpha a = 0$$

$$\text{but } \sin n\pi = 0$$

$$\alpha a = n\pi$$

$$\alpha = \frac{n\pi}{a}$$

$$\text{And } \alpha^2 = \frac{n^2 \pi^2}{a^2} \longrightarrow (10)$$

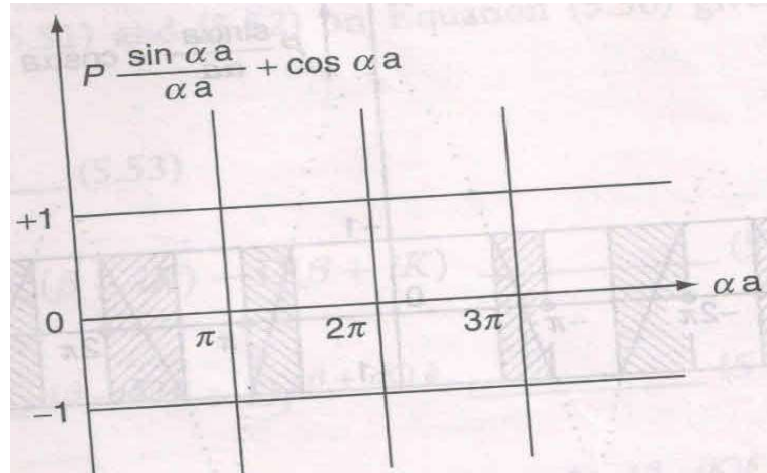
$$\text{But } \alpha^2 = \frac{8\pi^2 mE}{\hbar^2} \longrightarrow (3)$$

Equating the equations (10) and (3)

$$\frac{8\pi^2 m E}{h^2} = \frac{n^2 \pi^2}{a^2} \quad (\text{Since, } \hbar = \frac{h}{2\pi})$$

$$E = \frac{n^2 \hbar^2}{8ma^2}$$

This expression shows that the energy spectrum of the electron contains discrete energy levels separated by forbidden regions.



iv. When $P \rightarrow 0$, then Equation (9) becomes

$$\cos \alpha a = \cos Ka$$

$$\alpha = k$$

$$\alpha^2 = K^2$$

$$\text{But } \alpha^2 = \frac{2mE}{\hbar^2}$$

$$(\text{Since, } \hbar = \frac{h}{2\pi} \text{ and } k = \frac{2\pi}{\lambda})$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$= \frac{h^2}{2m} \left(\frac{1}{\lambda^2} \right)$$

$$= \frac{h^2}{2m} \left(\frac{p^2}{m^2} \right)$$

$$= \frac{p^2}{2m}$$

$$E = \frac{1}{2} m v^2$$

The equation shows that all electrons are completely free to move in the crystal without any constraints. Hence, no energy level exists i.e., all the energies are allowed to the electrons. This equation represents energy of a free particle.

